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REVIEW

A survey on social media popularity prediction: concepts, technologies, applications, and challenges

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Abstract

Due to the rapid development of user scale, interactive forms, and social media content diversity on social platforms, social media popularity prediction, aiming to predict the popularity of user generated content on social network according to multimodal and statistical features, is a significant challenging task, which is essential for many related downstream applications, such as personalized recommendation systems, marketing management, and public opinion governance, etc. Abundant research works provide a better understanding of the key factors driving the diffusion of social media content, but lacking an unified taxonomy of social media popularity prediction methods. In this paper, we comprehensively review social media popularity prediction methods over the past decade. Specifically, we first formally define the social media popularity prediction task and introduce the type of multi-modality content. Subsequently, we provide a novel taxonomy architecture from methodology perspective, which categorize the existing works into the five groups and their subclasses in each group, and analyze their advantages and disadvantages. Meanwhile, we outline the mainstream public datasets and evaluation metrics in detail. Finally, we discuss the open application and future direction in this field. This paper comprehensively provides valuable references for readers to understand the cutting-edge researches.

Keywords Social media popularity prediction · Deep learning · Multi-modality features · Large language model

1 Introduce

With the rapid development of mobile internet technology, many social media platforms (e.g., Facebook, Instagram, X, Weibo, and Weixin) have become important tools for modern communication, in which users share their idea, experiences, and knowledge, gradually becoming the main media venues for information dissemination and user interaction (Liu et al. 2025a, c). **U**ser **G**enerated **C**ontent (**UGC**) on social networks, also known as social media content, continues to grow exponentially daily, making it difficult to filter out the most relevant and attractive material. Due to the numerous and complex factors that affect popularity, **S**ocial **M**edia **P**opularity **P**rediction (**SMPP**) aims to evaluate the future popularity of social media content according to their semantic and statistical features, which has become a challenging and significant task (Xu et al. 2025b). Research on SMPP not only advances the understanding of the popularity mechanism of social media content, but also facilitates many downstream tasks, including but not limited to content optimization (Kleinberg et al. 2024), marketing strategies (Zahran 2025), public opinion monitoring (Mao et al. 2025),

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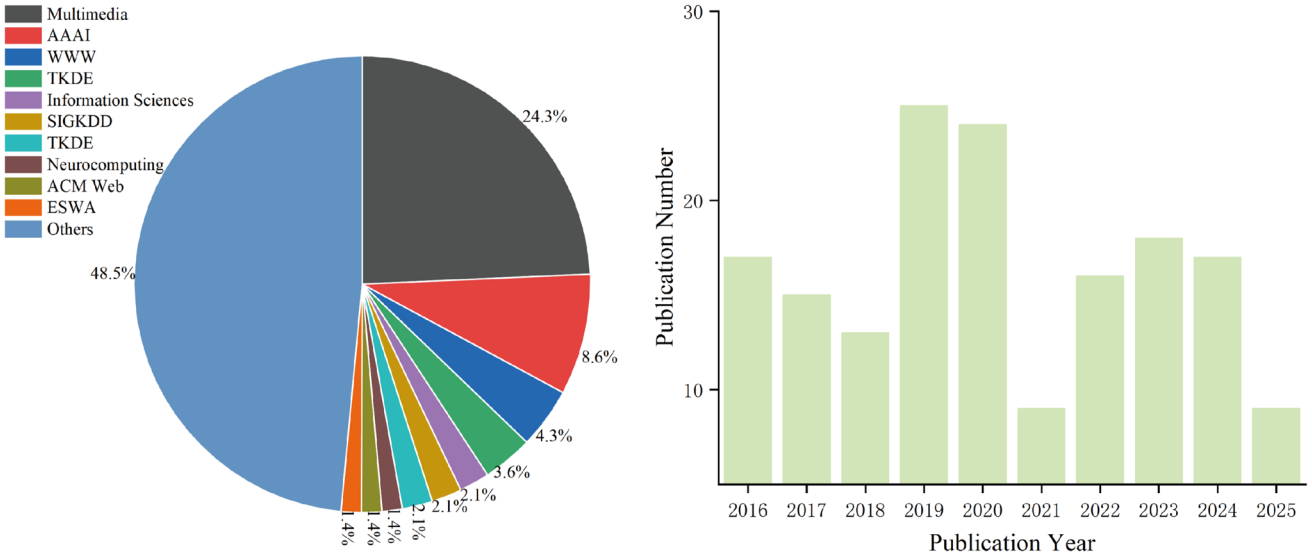


Fig. 1 Publication Distribution by Venue/Year

personalized recommendation systems (Li et al. 2024b), etc., attracting great attention from both academia and industry community.

Social media content is a type of multi-modal user generated data, including images (Hu et al. 2024; Cheng et al. 2024), text (Kim et al. 2024), videos (Cho et al. 2024; Xie et al. 2020), etc. Existing work discovers an interesting phenomenon that most UGC only receive little attention, while a few UGC attract a large number of user attention (Liu et al. 2025b). Thus, many researchers always invest considerable effort into excavating multi-modal features from social media content to analyze the popularity mechanism over the past decade, achieving satisfactory research, as shown in Fig. 1 (right). Meanwhile, existing works that are reviewed in this survey in international journals/conferences related to data mining, multi-modal feature analysis, time series analysis, etc., such as Multimedia, AAAI, WWW, TKDE, SIGKDD, CVPR, IJCAI, KDD, IPM, and KBS, etc., demonstrating that it is an interdisciplinary subject, as shown in Fig. 1 (left). Afterwards, to summarize these research findings, researchers attempt to summarize the existing methods from different perspectives in recent years, as shown in Table 1. For example, Liu et al. analyze the key factors that influence, feature extraction, and classification methods for social media popularity (Liu et al. 2022). Moreover, a more recent survey focuses mainly on deep learning methods (Fan 2025). However, they either fail to comprehensively review the existing methods or contain the latest literature, greatly limiting the development of this field. Besides, the definition of the social media popularity prediction task varies significantly between different tasks. Therefore, there is an urgent need to comprehensively review the latest social media popularity methods with a unified standards taxonomy, further promoting research of this field.

To address the above challenges, we comprehensively review existing efforts over the past decade. Specifically,

Table 1 Comparison Between Different Social Media Popularity Prediction Surveys (Generative Methods: **GM**, Feature Engineering Single Model Methods: **FESMM**, Feature Engineering Ensemble Model Methods: **FEEMM**, Time Series Methods: **TSM**, Multimodal Machine Learning Methods: **MMLM**, Multimodal Deep Learning Methods: **MDLM**, and Large Language Model Methods: **LLMM**)

Reference	Year	GM	FESMM	FEEMM	TSM	MMLM	MDLM	LLMM
Moniz and Torgo (2019)	2019	-	✓	✓	✓	-	-	-
Gao et al. (2019)	2019	✓	✓	✓	✓	-	-	-
Rathord et al. (2019)	2019	-	✓	-	-	-	-	-
Raghunathan et al. (2021)	2021	-	-	-	✓	✓	✓	-
Anand et al. (2022)	2022	-	✓	✓	-	-	✓	-
Liu et al. (2022)	2022	-	-	-	-	-	✓	-
Fan (2025)	2025	-	✓	✓	-	-	✓	✓
Ours	2026	✓	✓	✓	✓	✓	✓	✓

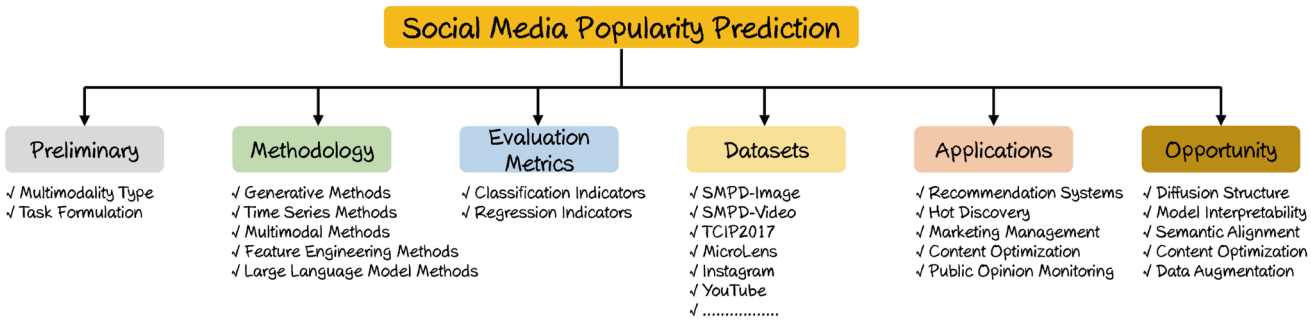


Fig. 2 The organizational structure of this paper

social media popularity prediction can be considered as either a classification task or a regression task. Then, according to feature extraction and learning strategies, we propose a novel taxonomy of social media popularity prediction methods from the methodology perspective, which is used to divide existing works into generative methods, feature engineering methods, time series methods, multimodal methods, and large language model methods. Meanwhile, we introduce the common public available benchmark datasets and evaluation metrics, such as classification indicators and regression indicators. Besides, we discuss the application and future direction in this field. Compared with previous research, the main contributions of this paper are summarized as follows.

(1) **Novel Taxonomy and Comprehensive Review.** According to feature extraction and learning strategies, we propose a novel taxonomy to divide existing works into five classes and their sub-classes over the past decade, and provide finer-grained analysis of trade-offs, advantages, and disadvantages of existing methods.
(2) **Resource Compilation.** We outline the latest and representative model performance on public datasets, and provide a detailed introduction to public dataset and the evaluation indicators, offering an intuitive reference in model comparison and evaluation for researchers.
(3) **Application and Opportunity Analysis.** We further discuss the application and present potential directions for future research, providing clear guidance for the advancement of this field.

The remainder of this paper is organized in the following structure, as shown in Fig. 2. Section 2 reviews the preliminary work that formally defines the social media popularity task and introduces the multi-modality types, which provides the foundation for the understanding of the rest. In Section 3, we briefly provide a comprehensive review of social media popularity methods, followed by publicly available standard benchmark datasets in Section 4. Section 5 presents the evaluation metrics, including the classification indicators and the regression indicators. The application of social media popularity and future direction are discussed in Sections 6 and 7, respectively. Finally, we conclude with this paper in Section 8.

2 Preliminary

In this section, we primarily define the social media popularity prediction problem and provide a mathematical expression. Meanwhile, we introduce the relevant types of social media content, which can facilitate understanding the rest of the paper.

2.1 Multimodality type

Users generate a large amount of multi-modal content for sharing their ideas on social networks, which mainly describes the collective term for multiple data signals from different physical forms, perception channels, or representations of the same thing or scene, including text, images, and videos, etc. Each modality possesses the unique characteristics and supplements semantic information of other modalities. For example, Text logically carries ideas, images visually evoke resonance, and videos dynamically create immersion. Multi-modal content on social networks not only expands the dimensions of information expression but also enhances the depth and breadth of information dissemination. Next, we will describe them in detail.

Text. As the most fundamental, efficient, common and important information carrier, this modality primarily conveys information, exchanges viewpoints, and expresses emotions with the combination of linguistic symbols (Zhang et al. 2021; Qian et al. 2022), such as letters, words, sentences, punctuation marks, etc.

Image. As the saying goes, "A picture is worth a thousand words". This modality refers to static two-dimensional content composed of visual elements (Hu et al. 2024; Abousaleh et al. 2020), such as color, lines, shape, light and shadow, composition, etc. Meanwhile, this modality provides an intuitive and fast way of emotional impact and information

transmission, which is a powerful complement for textual content.

Video. This modality, consisting of video frames, is a form of continuous content with a relatively short duration (from a few seconds to a few minutes) (Li et al. 2024a; Ding et al. 2018). This modality integrates dynamic visual images, audio, text, and other elements, etc., which have become the most immersive and expressive form currently available.

2.2 Task definition

According to previous research, social media popularity prediction (SMPP) is roughly considered as the regression task (Wang and Zhang 2017) and the classification task (Li et al. 2016), which constitutes the benchmark framework for the evaluation from the methodological level. For classification task, popularity is treated as a discrete value representing the popularity level (Cho et al. 2024; Jeon et al. 2020), which denotes whether the popularity exceeds the pre-setting threshold value, such as low, medium, and high levels. For the regression task, popularity is treated as an exact value (Tang et al. 2022).

Popularity score . Inspired by previous research (Wu et al. 2017), the popularity degree of social media content is measured with popularity scores, which consists mainly of retweets, comments, favorites, likes, and views, etc. To mitigate the significant disparity in popularity between different social media content, the logarithmic function is utilized to normalize the popularity values, the standardized popularity score is shown by the following (1).

$$p = log_2\left(\frac{r}{d} + 1\right) \tag{1}$$

Where r expresses the summation of retweets, comments, favorites, likes, and views, and d implies the number of days since user generated content is released.

Social media popularity prediction. Given user generated content $C = \{c_1, c_2, ..., c_N\}$, the object learns multi-modal features representations from social media content c_i by building a classification model or regression model $f(\cdot)$ to predict social media popularity $\hat{p}$, including the negative popularity and positive popularity, where c_i denotes the multi-modal content $i - th$ that contains image, text, video, and user profiles, which is formally defined as shown the following (2).

$$\hat{p} = f(T, I, V, U, \Theta) \tag{2}$$

Where Θ denotes the model parameters, $\hat{p}$ is the predictive value, such as the discrete popularity level for the classification task or the exact value for the regression task, $f(\cdot)$ represents the proposed regression model or the classification model, T represents the text modality, V signifies the video modality, I refers to the image modality, and U represents user profiles.

3 Methodology

In this section, inspired by previous works (Bahi et al. 2026), according to the feature extraction mode of multi-modality content on social media, model types, and training strategies, existing efforts are principally divided into five subcategories: **Generative Methods**, **Feature Engineering Methods**, **Time Series Methods**, **Multi-Modality Methods**, and **Large Language Model Methods**, as shown the Fig. 3. Next, we will introduce them in detail.

3.1 Generative methods

This type of method based on the Hawkes point process proposed by a British statistician (Hawkes 1971), a famous stochastic point process model, is the classical representative of generative methods. In light of the observed events, these

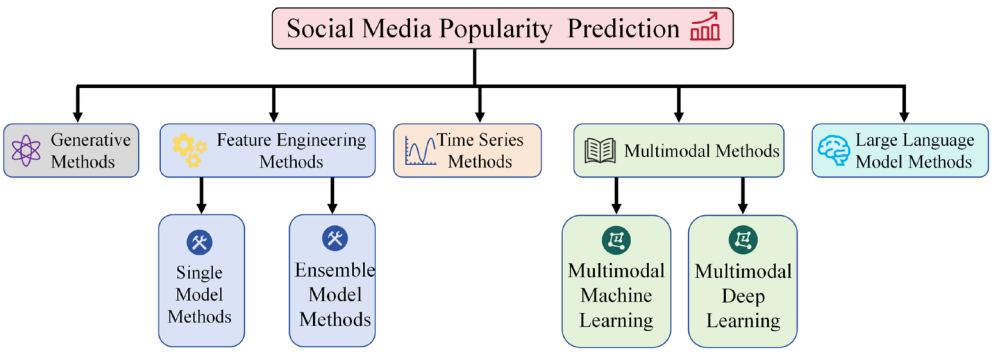


Fig. 3 The taxonomy of social media popularity prediction in this paper

models consider the series of event occurrence phenomena as parameterized models, and then adjust the model parameters. They rely on the fact the occurrence of historical events potentially increases the instantaneous probability of new events occurring, which predominantly lies in the determining conditional intensity function by depending on time and the historical events occurrences, providing a precise quantitative framework for understanding the chain reactions in the real world. The base intensity is shown as the following (3).

$$\lambda(t) = \mu + \sum_{t_i < t} \phi(t - t_i) \tag{3}$$

Where μ implies the basic intensity, t_i is the time of new event occurrences, $\lambda(t)$ represents the intensity function of new event occurrences, and $\phi(\cdot)$ symbolizes the trigger kernel.

On top of the classical Hawkes process, many researchers subsequently propose various variant models by designing different conditional intensity functions, which are used to simulate and verify the impact of different key factors on social media popularity prediction. For example, Mishra et al. (2016) design a novel Hawkes process with a self excitation mechanism, effectively estimating the importance of content virality, memory decay, and user influence. Kobayashi and Lambiotte (2016) construct a time-dependent Hawkes process model for studying the temporal patterns of retweet, which not only fully considers the retweet patterns and the decay mechanism of information dissemination, but also designs a parameter optimization method to predict popularity at different time scales. To fill in the blank in practical tools, Kong et al. (2018) present an interactive visualization system tool based on the Hawkes process, which intuitively analyzes and displays the dissemination and popularity of online videos by retrieving information from the video metadata. Moreover, to analyze group behavior patterns on social networks, Bao et al. (2019) first propose a novel Hawkes process model with the exploratory adaptive peeking window to filter historical data, which predicts the final popularity of social media content based on initial observation data, improving predictive accuracy. Comprehensively considering social behavior and network characteristics, such as follower counts and activity variation observed in collective re-sharing behavior, Zadeh and Sharda (2022) introduce a flexible multivariate Hawkes process model to analyze the impact of information diffusion patterns on popularity prediction, which is the superposition of univariate point processes of multiple different types, the conditional intensity are formulated as shown the following (4).

$$\lambda_i(tH_t) = \lambda_{B_j}(r,t) + \sum_{k=1}^{d} \rho_{kj} \int_0^t h_{kj}(t-s)\, g_j(x)\, dN_j(s) \tag{4}$$

Where $j = 1, 2, 3, ..., d$, ρ_{kj} is the total intensity of the offspring, $N_j(t) = [N_1(t), N_2(t), ..., N_d(t)]$, $g_j(\cdot)$ refers to the boost functions, $\lambda_{B_j}(\cdot)$ presents the background intensity.

Discussion With the self motivating properties, this type of methods directly maps the core mechanism of historical events triggering future events occurrence to the viral behaviors (e.g, retweet, like), thus simulating the explosive process of social media popularity. Meanwhile, they can not only predict the complete trajectory of popularity evolution and generate the specific series of interactive events, but also express the characterization of the information life cycle and explosion patterns for micro-level. Furthermore, these methods have strong interpretability and mathematical theoretical support. For example, the base intensity reflects the initial attractiveness of the social media content, the trigger strength represents the ability to transmit viruses, and the decay rate corresponds to the duration of public interest. However, they primarily focus on the dissemination dynamic of social media content under strong assumptions from prior knowledge, and ignore the impact of multimodal content on social media(e.g, text, image, video), thus they overly simplified the models, which is not line with the real scene, further limiting their application scope.

3.2 Feature engineering methods

Initially, features engineering methods are the mainstream approach for social media popularity prediction, which manually extract the effective multi-modal features from the specific social media content associated with the metadata information (e.g. the spatiotemporal characteristics (Lai et al. 2020), image feature (Wang and Zhang 2017), topic (Purba et al. 2020), social characteristic (Keneshloo et al. 2016), numerical feature (Li et al. 2018), text feature (Li et al. 2018), emotions feature (Purba et al. 2020), video characteristic (Sibo et al. 2021), tag (Chen et al. 2020), and user profiles (He et al. 2019), and then the handcrafted features are fed into the regression model or the classification model for predicting social media popularity, such as support vector regression model (Gupta and Yang 2019), the Catboost algorithm (Kang et al. 2019), the gradient boosting regression tree algorithm (Wang and Zhang 2017), the LightGBM algorithm (He et al. 2019), the random forest algorithm (Purba et al. 2020), the XGBoost algorithm (Nisa et al. 2021), the multiple linear regression algorithm (Sibo et al. 2021), the matrix factorization methods (Lv et al. 2017), the ridge regression algorithm (Ziyada and Shamoi 2024), and the decision tree algorithm (Zhu et al. 2022), etc. According to the number of social media popularity predictors, this type of method is divided into **the single model methods** and **the ensemble model methods**, as shown the Fig. 4 and the Table 2.

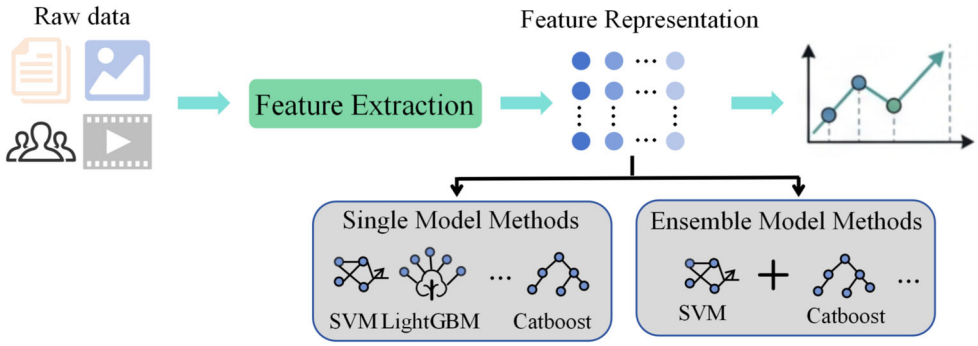


Fig. 4 The taxonomy of feature engineering methods

3.2.1 Single model methods

According to the task definition, this type of method employs a single traditional machine learning approach for predicting social media popularity, which has garnered increasing attention due to their advantages in computational efficiency, such as the regression model and the classification model. For example, on top of the classical support vector model, Trzciński and Rokita (2017) further advance a support vector algorithm based on the gaussian radial basis functions for the videos popularity predict, as shown in the following (5).

$$p = \sum_{k=1}^{K} \alpha_k \bullet \Phi\left(X(v, t_r), X(k, t_r)\right) + b \tag{5}$$

Where $\Phi(x, y) = exp(-\frac{\|x-y\|^2}{2\sigma^2})$ expresses the gaussian radial basis function with the parameter σ, $X(v, t_r)$ implies the feature vector of video v at time t_r, b is the intercept and α_k is the weight coefficient. Huang et al. (2018)

Table 2 Summary of feature engineering methods for social media popularity prediction.(R:Regression; C:Classification)

Reference	Year	Task	Methods	Features
Keneshloo et al. (2016)	2016	R	Multi Linear Regression	Metadata, Context, Time, Social
Lv et al. (2017)	2017	R	Linear Regression, Support Vector Regression, Matrix Factorization	Visual, Social, Time
Wang and Zhang (2017)	2017	R	Ridge Regression, Gradient Boosting Regression Tree	Metadata, User, Length of Title, Time
Trzciński and Rokita (2017)	2017	R	Support Vector Regression	Visual, Time
Li et al. (2018)	2018	R	LightGBM	Numerical, Metadata
Huang et al. (2018)	2018	R	Random Forest Regression	Post, User
Li et al. (2016)	2019	C	Concept Drift Model	Diffusion Features, Metadata
He et al. (2019)	2019	R	LightGBM	User, Numerical, Text
Kang et al. (2019)	2019	R/C	Catboost	Categorical, Numerical, User
Purba et al. (2020)	2020	R	Random Forest	User, Post, Keyword, Hashtag, Engagement, Sentiment
Carta et al. (2020)	2020	C	Gradient Boosting	likes, Time , Text
Lai et al. (2020)	2020	R	CatBoost	Image, Category, Space-time, User Profile, Tag
Wang et al. (2020)	2020	R	CatBoost	Text, Visual, Numerical, User
Chen et al. (2020)	2020	R	XGboost	Content, Social, Global features
Nisa et al. (2021)	2021	C	XGBoost	Category, Definition, Duration, Score, Views
Sibo et al. (2021)	2021	R	Multiple Linear Regression	Channel, Time
Zhu et al. (2022)	2022	C	K-SC Clustering, Decision Tree	Publisher, Text, Content
Amrutha (2022)	2022	R	Support Vector Regression	Views, Social, Visual
Ziyada and Shamoi (2024)	2024	C	Ridge Regression	Video, Sentiment, Emotion

reduce dimensionality and complete missing data on metadata information and user features of social posts with binary encoder and imputation strategies. Kang et al. (2019) apply the Catboost algorithm to enrich the feature space by introducing user information, and design a pseudo label based data augmentation strategy, achieving satisfactory results. Meanwhile, to solve the generalization problem caused by data loss and the potential short term dependencies between posts, Wang et al. (2020) learn the features of social media content by combining the CatBoost algorithm and the sliding window averaging strategy, improving the prediction accuracy. Moreover, Carta et al. (2020) analyze the importance of metadata information from social posts with the XGBoost algorithm. Bohra and Bhatnagar (2021) combine graph clustering and tensor decomposition methods for popularity prediction at the group level, and Adam is applied to optimize the objective function. Then, to recommend post tags and improve its popularity, Chakrabarti et al. (2023) built a social influence framework based on topic tags that considers the popularity of users and posts, significantly improving the robustness and reliability of the model. In addition, Jeong et al. (2024) discuss the effect of metadata information on social media popularity prediction by using machine learning methods, such as the mixed linear model, the support vector regression model, the random forest algorithm and the XGBoost algorithm.

3.2.2 Ensemble model methods

This type of method employs an ensemble framework consisting of multiple machine learning methods (e.g. the regression model or the classification model) for social media popularity prediction, which receive a great deal of research attention from the academic community, benefiting from the fact these methods can effectively combine the strengths of each approach. For example, Zhu et al. (2022) propose an ensemble framework to explore the patterns of topic popularity changes with the time series clustering model and the classification model, in which the time series clustering algorithm is identified the fundamental types of topic popularity without relying on preexisting popularity data, and then the decision tree model is utilized to predict popularity. Hidayati et al. (2017) learn user characteristics, post metadata, and image esthetic features with the ensemble model for social image popularity prediction, its cluster centroid is shown by the following (6).

$$\mu_k = argmin \sum_{X_i \in S_k} \sum_{X_j \in S_k} \left(X_i - X_j\right)^2 \tag{6}$$

Where S_k denotes the data set of the cluster k, X_i and X_j are the sequence i and j, respectively. Similarly, Huang et al. (2017) analyze user profiles, visual content, and contextual information with an ensemble model for social image popularity prediction. To further enhance complementarity and predictive performance, Lin et al. (2019) reduce the dimensionality and select features with the random forest algorithm, and then the semantic and social characteristics of the image are learned using a hierarchical deep stacking framework consisting of different regression models. Zhang and Bao (2020) explore the impact of randomness, dynamism, and uncertainty of social networks on popularity prediction with multiple traditional time series models, which adapt to the dynamic changes of the evolutionary model and improve model performance due to dynamically combining sub models with the weighted parameters, as shown by the following (7).

$$\omega_{i,j} = \frac{exp\left(-\left(\hat{v_{i,j}} - v_{i,j}\right)^2\right)}{\sum_{i=1}^{3} exp\left(-\left(\hat{v_{i,j}} - v_{i,j}\right)^2\right)} \tag{7}$$

Where $\hat{v_{i,j}}$ expresses the predictive value of the model i within the time interval j, and $v_{i,j}$ implies the true value of the model i within the time interval j. Li et al. (2016) refined a popularity predictive model based on concept drift, which dynamically determines the weights of the classification models training on the datasets from the multiple different periods. Halim et al. (2022) identify and rank the important attributes of independent features (e.g., likes, comments, and downloads) with feature extraction methods, such as artificial neural networks, the k-nearest neighbor algorithms, and the support vector machine.

Discussion Due to the clear semantic meaning of the multimodal features selected manually, this type of method not only has strong interpretability of multimodal features and engineering practicality, but also requires low computational cost in the model training and inference processes; thus, they can meet the real-time requirements of online systems. Meanwhile, they can flexibly integrate multi heterogeneous features from social media content, user profiles, and social characteristics for social media popularity prediction. Furthermore, ensemble model methods can effectively alleviate the shortcomings of single models. However, this class of methods manually extracts the multimodal features of social media content from specific fields, which severely depends on the specific expertise experience and is a time-consuming, labor-intensive, and knowledge intensive project, thus these models lack generalization ability and are difficult to generalize to other domains. Meanwhile, the handcrafted features remain only at the statistical level, which fail to capture the deep semantic associations and complex patterns of social media content, which are essentially static and limited, limiting the model performance.

3.3 Time series methods

Generally, social media popularity, considered as a time series prediction task, is stored in the form of numerical sequences that change over time (Song et al. 2023). The core idea is to forecast the future popularity of social media content based on historical data. Therefore, researchers commonly employ sequential deep learning models for predicting social media popularity, such as regression models (Trzciński et al. 2017; Guan et al. 2025), LSTM (Zhang et al. 2021; Trzciński et al. 2017), and the attention mechanism (Wang et al. 2018; Hsu et al. 2022), etc., as shown the Table 3. For example, to explore the temporal characteristics, Wu et al. (2017) propose a deep temporal context network framework, which mainly analyzes the contextual sequence and temporal correlations with the temporal context modules and the multi-scale temporal attention mechanism. Meanwhile, Nguyen et al. (2019) propose a non-recursive neural network with an attention mechanism framework, which can autonomously filter the relevant inputs and capture long-term dependencies between the sequences by using two stage attention mechanism. Subsequently, Sangwan and Bhatnagar (2021) explore the influence of video content and temporal features with the bidirectional long short term memory model, in which videos are treated as temporal data rather than static image collections. To analyze the impact of external factors, metadata information and temporal features on popularity prediction, Liu et al. (2023a) devise an adaptive temporal knowledge graph model considering temporal features and knowledge graph information, which captures temporal dynamic features and complex relationships between entities with attention mechanisms and the relationship graph convolutional network, respectively, the optimization objective loss is shown by the (8).

$$Loss = \sum_{i \in \mathcal{D}} \sum_{t \in [1,n]} MSE\left(y_t^i, \hat{y}_t^i\right) \tag{8}$$

Where $\mathcal{D}$ denotes the video data set, t is the predictive time, and y_t^i and $\hat{y}_t^i$ symbolizes the actual value and the predictive value, respectively. Moreover, Yang et al. (2024) measure the intensity of information interaction and the similarity of time series with dynamic time warping algorithm and information entropy, and then integrate the temporal features with the recurrent neural network, the information entropy is applied as shown by the following (9).

$$H(x) = -\sum_{n}^{i=1} p(x_i) \, log_b p(x_i) \tag{9}$$

Where $p(x_i)$ symbolizes the probability of the occurrence of the $i-th$ social media content, $H(x)$ presents the value of the information entropy and b determines the unit of the information entropy $H(x)$. Subsequently, Sangwan and Bhatnagar (2025) further propose a novel multi-branch long short term memory framework, which maps video features into low-dimensional space for video popularity prediction.

Discussion This type of method considers social media popularity as sequential numerical data, accurately depicting the dynamic evolution of social media popularity. Meanwhile, they efficiently analyze the long-term trends and cyclical patterns of sequential social media popularity. Furthermore, such models can meet the needs of real-time prediction due to their fast training speed and low complexity. However, these methods fail to consider the complicated semantics of multimodal features from social media content (e.g., metadata information, text, image, video), which cannot be obtained solely from the sequential numerical popularity, thus ignoring the importance of social media content. Besides, the model performance is highly dependent on the continuity and quality of the sequential numerical popularity. Therefore, this type of method is more suitable for the short-term and regular popularity prediction.

Table 3 Summary of time series methods for social media popularity prediction.(R:Regression; C:Classification)

Reference	Year	Model	Task	Methods
Wu et al. (2017)	2017	DTCN	R	LSTM, Attention Mechanism
Trzciński et al. (2017)	2017	Popularity-LRCN	C	CNN, LSTM
Wang et al. (2018)	2018	UMAN	R	Attention Mechanism, LSTM, CNN, One-hot
Nguyen et al. (2019)	2019	DA-RNN 5	R	Attention Mechanism, Modified Transformer
Sangwan and Bhatnagar (2021)	2021	-	R	BiLSTM
Zhang et al. (2021)	2021	DeepBlue	R	Bi-layer LSTM, Poisson model
Liu et al. (2023a)	2023	ATKN	C	ALSTM, AFM, RGCN
Yang et al. (2024)	2024	EEMD-MLSTM	R	Information Entropy, MLSTM
Guan et al. (2025)	2025	FPF-SR	R	SSC, Fourier Transform
Sangwan and Bhatnagar (2025)	2025	MLEF-DL	R	LSTM, CNN

3.4 Multimodal methods

Generally, social media content generated by netizen on social networks is typically multi-modal consisting of image (Zhong et al. 2024), text (Zhang et al. 2023), videos (Jing et al. 2017), etc., which not only contains rich semantic and emotional information (Liao et al. 2019; Zhang et al. 2020), but also expands the depth of information understanding through cross modal complementarity and enhancement (Hu et al. 2024). Recently, the large language model and the pretrained model have attracted significant attention in feature extraction applications, which mainly transform the effect features extracted from multi-modal content into dense vectors, such as visual language model (Lin and Lee 2024; Zhang et al. 2024b) and visual models (Cho et al. 2024). Therefore, many multi-modal methods are proposed to predict social media popularity (Zhang et al. 2018b; Xu et al. 2020), following the "**Feature Extraction–Feature Learning**" paradigm. According to the characteristics of feature learning process, these methods are roughly divided into the **multi-modal machine learning** and the **multi-modal deep learning**, as shown the Fig. 5 and the Table 4.

3.4.1 Multimodal machine learning

This type of method inputs the extracted multi-modal features into traditional machine learning for social media popularity prediction, such as the CatBoost algorithm (Tan et al. 2022), the XGBoost algorithm (Li et al. 2017), etc. According to the number of feature learning methods, these methods are mainly categorized into two subcategories: the **Multi-modal Single Model Methods** and the **Multi-modal Ensemble Model Methods**.

Multi-modal single model methods This type of method comprehensively combines the advantages of pretrained models and traditional machine learning for social media popularity prediction. For example, Aloufi et al. (2017) analyze the impact of key factors on popularity with the ranking support vector machine, such as visual content, social characteristic, and textual information, etc. Jing et al. (2017) innovatively propose the low rank representation based on multiview learning framework for the micro videos popularity prediction, which fuse the multiview features with the shared low rank subspaces, effectively alleviating the regularization of the graph. Moreover, Li et al. (2017) input the extracted high-level features into the XGBoost algorithm for social media popularity prediction. Hsu et al. (2019) jointly apply the encoder-decoder based on the CNN-RNN framework, the word to vector and one-hot encoder to effectively extract features from image, text and metadata information, which are fed into the LightGBM algorithm for popularity prediction, demonstrating that image semantics and textual features are vital. Furthermore, Chen et al. (2019b) combine a collaborative expert based on the XGBoost algorithm framework, which learns the fusion of embedding representation from the video and text modality, the combining process is shown in the following (10).

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \; f_k \in \mathcal{F} \tag{10}$$

Where $\mathcal{F} = \omega_{q(x)}$ is the search space for the regression tree sample, $q : R^m \longrightarrow T, \omega \in R^T$, q maps the sample to

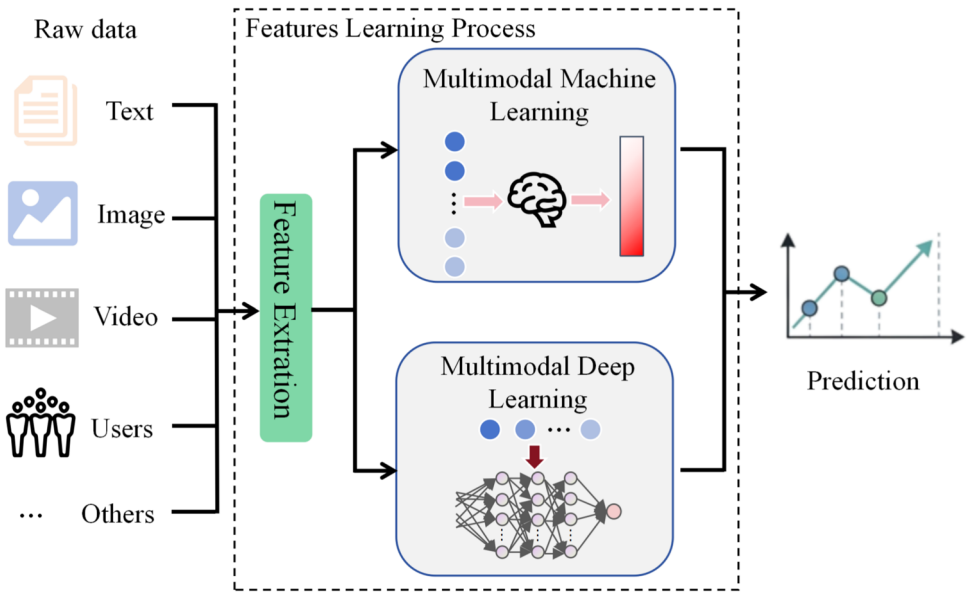


Fig. 5 The taxonomy of multimodal methods

Table 4 Summary of multimodal methods for social media popularity prediction

Ref.	Year	Model	Task	Methods	Modal Extraction	Modal
Li et al. (2017)	2017	-	R	XGBoost	CNN	I
Jing et al. (2017)	2017	TLRMVR	R	Least Squares Regression	DeepSentiBank, ImageNet, CoreNLP, Sentence2Vec	V
Zhang et al. (2018b)	2018	Dual-attention	C	LSTM	ResNet50, LDA, Word Embedding	T/I
Zhang et al. (2018a)	2018	UHAN	R	Attention Mechanism, LSTM	VGGNet, One-hot	T/I
Hsu et al. (2019)	2019	-	R	LightGBM	CNN-RNN, Word to Vector, One-hot	T/I
Ding et al. (2019)	2019	-	R	Dense Layer	ResNet-101, BERT	T/I
Xu et al. (2020)	2020	-	R	Attention Mechanism, LSTM	ResNet50, Word2Vec	T/I
Su et al. (2020)	2020	FDTM	R	SVM	AlexNet, CoreNLP, Sentence2Vector	V
Abousaleh et al. (2020)	2020	VSCNN	R	PCA, CNN	LBP, GIST, VGG19	I
Lin et al. (2021)	2021	-	R	Attention Mechanisms, MLP	ResNet50	I
Wu et al. (2022)	2022	DEVL	R	CatBoost	Clip, Geocoder, BERT, Nima, HyperIQA	T/I
Tan et al. (2022)	2022	-	R	CatBoost	Transformer, BERT, Fasttext, GloVe	T/I
Mao et al. (2023)	2023	ECSF	R	CatBoost	CLIP, Glove	T/I
Hsu et al. (2023)	2023	MFTM	R	LightGBM, TabNet, PCA	BLIP, Geocoder	T/I
Chen et al. (2023)	2023	DFT-MOVLT	C	MLP	ALBEF	I
Liu et al. (2023b)	2023	SMPC	R	CatBoost	Word2vec, One-hot, PIL, IF-IDF, SVD	T/I
Arazzi et al. (2023)	2023	TweetGage	C	GCN, MLP	Bert	T
Hu et al. (2024)	2024	DSPT	R	LightGBM, CatBoost	CLIP, RoBERTa	T/I
Lin and Lee (2024)	2024	MMF	R	Contrastive Learning, Attention Mechanism	Swin, MiniGPT4, Bert	T/I
Cho et al. (2024)	2024	AMPS	C	Attention Mechanism, LSTM	ResNet50, Bert	I/V
Kim et al. (2024)	2024	Multi-Pop	C	LSTM	DenseNet201, RoBERTa	T/I
Ye et al. (2025a)	2025	-	R	CatBoost, TabNet, MLP	CLIP, Word2Vec, SVD	I
Ye et al. (2025b)	2025	MVP	R	CatBoost, PCA	XCLIP, Word2Vec	V
Cheng et al. (2025)	2025	ICPF	R	Attention Mechanism, VIT	CLIP	T/V
Xu et al. (2025a)	2025	SKAPP	R	L-GNN, V-GNN, Attention Mechanism	GraphAdapter	T/I

each tree structure with the indexed leaf nodes, T implies the number of leaf nodes in each tree, ω is the weight of the leaf nodes in each tree and $f_k(\cdot)$ denotes the independent tree with the structure and weights of the leaf nodes. Meanwhile, Su et al. (2020) provide the normalized laplacian matrix fusion to integrate heterogeneous modal features from visual, auditory, social and text, enhancing model performance, as shown in the following (11).

$$L(S_t) = \sum_{k=1}^{K} \beta_k L(S_{kl}) \tag{11}$$

Where $\sum_{k=1}^{K} \beta_k = 1$, $\beta_k \geq 0$ symbolizes the weighted parameter, and $L(S_l)$ refers to the laplacian matrix of $l-th$ modality subspace. In addition, Wu et al. (2022) study the impact of multi-modal semantics, image quality, and scene on popularity prediction by extracting effective features from the image. Mao et al. (2023) innovatively integrate high-order features from text and image with the stacked feature method, significantly improving predictive accuracy. Additionally, Liu et al. (2023b) achieve multi-dimensional feature collaborative learning with three different title modeling mechanisms, and then combine the polymerized features with the Catboost algorithm for popularity prediction. Besides, Liu et al. (2023c) input the structural relationships extracted from the image with the pretrained model into the Catboost algorithm for popularity prediction.

Multi-modal ensemble model methods This type of method, named the multimodal ensemble model methods, primarily combines the powerful semantic extraction capabilities of pre-trained models based on deep learning and the controllable characteristics of traditional machine learning methods, which relieves the generalization error easily caused by the single method, significantly improving the predictive accuracy. For example, Arora et al. (2023) analyze the complex relationship between multi-modal features and popularity with an ensemble approach, including text, images, metadata information, emotions, etc., which consists of several traditional machine learning methods. Meanwhile, Hsu et al. (2023) apply the LightGBM algorithms and the TabNet algorithms to further investigate the impact of different feature combinations, such as identity features, external features, and metadata information, etc. Chen et al. (2023) enhance the understanding ability of the visual language model towards posts by introducing numerical features, and further fine-tune the ensemble method consisting of the regression model and the weighted classification model, improving model performance. Tu et al. (2024) jointly utilize the CatBoost algorithms and LightGBM algorithms to predict social media popularity, which achieve the seamless integration of visual and textual modality with the aligned semantic space, and the dynamic fusion strategy is shown by the following (12).

$$p = \lambda \times p_{Catboost} + (1-\lambda) \times p_{LightGBM} \tag{12}$$

Where λ implies the weighted parameter, $p_{Catboost}$ and $p_{LightGBM}$ are the predictive values. Moreover, Ye et al. (2025a) learn multi-modal features from visual, textual, and metadata information, etc, which contains the CatBoost algorithms, the TabNet algorithms, and the multi-layer perceptrons. Besiedes, Ye et al. (2025b) employ the pre-trained visual-language model to extract and analyze video features, user profiles, and contextual content, which are fed into the K-Fold CatBoost algorithm for social media popularity prediction. The Huber loss function is considered as the regression objective, as shown in the following (13).

$$Loss = \begin{cases} \frac{1}{2}(y-\hat{y})^2, & if\ |y-\hat{y}| \leq \delta \\ \delta|y-\hat{y}| - \frac{1}{2}\delta^2, & otherwise \end{cases} \tag{13}$$

Where y implies the actual value, $\hat{y}$ symbolizes the predictive value, and δ is the threshold. In addition, Liu et al. (2025d) extract the characteristics of the video modality with the different video classification architectures and encode the video content with the pretrained Bert model, demonstrating that the XGBoost algorithm and the neural network method achieve the best results.

Discussion This method effectively combines the powerful representation ability of deep learning with the efficient stability of traditional machine learning, in which they not only automatically extract rich semantic features from social media content, but also improve training and inference efficiency with the traditional machine learning method. Meanwhile, they do not require expensive computing power or massive annotated data to train or fine tune pre-trained models, significantly reducing computational costs. Furthermore, they also can analyze the importance of multi-modal features and flexibly integrate other algorithms to realize plug and play functionality, enchancing the model interpretability. However, these methods may lose the fine-grained information of multimodal features with pre-trained models, which benefits from the high compression of multimodal information. Meanwhile, they separate the multimodal feature extraction and multimodal feature semantic learning processes, easily resulting in the accumulation of errors and limiting model performance.

3.4.2 Multimodal deep learning

Inspired by the powerful implicit feature learning capabilities of deep learning (Liu et al. 2025c, a), researchers develop

an end-to-end learning framework to address the limitation of multi-modal machine learning methods. This type of method first extract multi-modal features from social media content by using deep learning methods or the pretrained models, which are fed into the deep learning models for social media popularity prediction, such as the multilayer perceptron (Ding et al. 2019), the attention mechanism (Li et al. 2024a; Lin et al. 2021), Graph Neural Network (GNN) (Arazzi et al. 2023), Transformer (Chen et al. 2022), the long short-term memory (LSTM) (Kim et al. 2024; Cho et al. 2024), Bi-LSTM (Bielski and Trzcinski 2018), the convolutional neural network (CNN) (Shin et al. 2020), the recurrent neural network (RNN) (Chen et al. 2019a), the hypergraph neural network (Cheng et al. 2024), the contrastive learning paradigm (Lin and Lee 2024), and the retrieval augmentation strategy (Cheng et al. 2025; Ji et al. 2023), etc. For example, Zhang et al. (2018a) introduce a novel user guided hierarchical attention framework for image popularity prediction, which hierarchically integrates the unstructured features of visual and textual modality. Abousaleh et al. (2020) excavate the visual and social characteristic with two independent convolutional neural networks, achieving satisfactory results. Xie et al. (2020) boast a multi-modal variational encoder-decoder framework to analyze uncertainties and noise during the feature learning process, and fuse multi-modal features with the bayesian algorithm, the object loss is shown by the following (14).

$$loss = \mathbb{E}_{p(m,y)}\left[\mathbb{E}_{p(z|m)}\left[logq\left(y|z\right)\right] - \lambda \cdot KL\left(p\left(z|m\right)||p\left(z\right)\right)\right] \tag{14}$$

Where z implies the hidden vector, $p\left(z|m\right)$ and $p(y|z)$ are the encoder and decoder distributions, m denotes the initialization features, y symbolizes the output vectors and $p(z)$ presents the hidden vector distribution. Furthermore, Mannepalli et al. (2023) further apply an adaptive optimization algorithm to fine-tune the parameters of long-term memory. Ultimately, Wang et al. (2023) reconstruct the multi-modal features from social media content with the hierarchical fusion mechanism, achieving satisfactory results. Zhang et al. (2024a) establish an innovative temporal hypergraph transformer framework, which fully captures the correlations between instances and temporal dependencies. Moreover, Zhang et al. (2024b) further analyze the impact of content relevance, user identity, and user influence similarity factors with three contrastive learning strategies, the total objective loss is shown in (15).

$$\mathcal{L}\left(\Theta\right) = \lambda \cdot \mathcal{L}_{reg}\left(\Theta\right) + \left(1-\lambda\right)\sum_{i}\alpha_i \cdot \mathcal{L}_{contra_i}\left(\Theta_i\right) \tag{15}$$

Where $\mathcal{L}_{reg}\left(\Theta\right)$ presents the loss of the predictive task, $\mathcal{L}_{contra_i}\left(\Theta_i\right)$ implies the loss of three discrimination tasks, λ and α are the weighted parameter for adjusting the loss of the task and contrast loss. In addition, Xu et al. (2025a) design a meta-retrieval strategy and a visual-language graph neural network to intelligently filter and integrate user generated content, the relevance score is calculated as shown in the following (16).

$$RS(c_i, c_j^D) = \sum_{f \in Q_i} IDF(c_i, f) \cdot \frac{T(c_{i,f}, c_{j,f}^D) \cdot (k_1+1)}{T(c_{i,f}, c_{j,f}^D) + k_1} \tag{16}$$

Where k_1 expresses the hyperparameter of the model, $T(c_{i,f}, c_{j,f}^D)$ measures the similarity of the feature $c_{i,f}$ and $c_{j,f}^D$, $IDF(c_{i,f})$ corresponds to the inverse frequency of the feature $c_{i,f}$, and Q_i is the query information.

Discussion This type of method is an end to end framework, which can automatically extract and learn multimodal features to predict social media popularity without the need for manual intervention, the semantic features extracted with the pre-trained model provide information initialization representations for social media popularity prediction. Besides, these methods can learn the non-linear relationships among multi-modal features with deep learning in the feature learning process, which effectively resolve the disconnect between feature extraction and task prediction, achieving global parameter optimization with the backpropagation algorithm. However, they require not only massive datasets from social networks due to the supervised learning mechanism, but also the computational costs for model training and inference processes due to the complexity of the model, easily leading to overfitting and weak generalization. Meanwhile, these model parameters are difficult to adjust, which is a time-consuming and laborious process. More importantly, the model interpretability is poorer, which makes it difficult to explain the causality relationship between the multi-modal features and the predictive results.

3.5 Large Language Model Methods

Recently, with its powerful natural language understanding and context generation capabilities, deep modeling contextual semantic, and efficient knowledge extraction on large-scale unlabeled data, large language models (LLM) have been widely applied in various domains, such as the intelligent content generation area, the cross linguistic information retrieval, and the dialog systems, etc. Researchers attempt to analyze social media popularity prediction with the large language models. For example, Rogoz et al. (2024) first construct a new Romanian language posts data set collected from Redis. Then, compared to the competitive models, the large language model achieves satisfactory results, which provides

the direction of the future studies. Moreover, to effectively alleviate feature dependence, insufficient generalization ability, and poor performance in cold start environments, Wang et al. (2024) propose a novel large language model instruction fine-tuning and human alignment for social media popularity prediction, which is fine-tuned with the low rank adaptation fine-tuning technology and the freeze fine-tuning technology, shown by the following (17). In addition, this framework mitigates the instability and non-convergence of human feedback with the improved direct preference optimization algorithm.

$$h = W_0 x + \triangle W x = W_0 x + BAx \tag{17}$$

Where B and A denote the low rank matrices, x expresses the input features, h is the output feature, and $\triangle W = BA$. Inspired by the contextual perception ability of large language models, Kayal et al. (2025) convert frame level visuals into serialized text description with the visual-language model, effectively solving the multi-modal semantic gap. Then, this model parses the interpretable titles and descriptive information with the large language model, as shown by the following (18).

$$t_i = VideoLLava\,(f_i w) \tag{18}$$

Where w expresses the visual context at the frame-level, t_i is the descriptive information generated, and $VideoLLava(\bullet)$ means the visual language model. Meanwhile, Chen et al. (2024) convert the information from the social media metadata into improved semantic text with the large language model and low rank adaptive technology, thus providing more narrative and intensive knowledge for the context of social media, low rank is shown by the following (19).

$$Attention\,(Q, K, V) = softmax\left(\frac{(Q + LQ)\,(K + RK)^T}{\sqrt{d_k}}\right)(V + RV) \tag{19}$$

Where L and R present the low rank matrices, Q, K, and V refer to the query matrix, key matrix, and value matrix, and d_k is the feature dimension.

Discussion With the excellent semantic parsing and context generation capabilities of the large language model, this type of method brings significant paradigm innovation for social media popularity prediction. These methods not only accurately capture implicit emotions, ironic intentions, and contextual information from the text modality, but also collaboratively analyze image/video content, comprehensively obtaining semantic features from social media context. Meanwhile, they have strong contextual inference and the potentiality of small sample learning, which perform causal inference on the relevance between posts and hot events through prompt mechanism. However, these methods have high computational cost and inference latency, severely limiting their feasibility in engineering deployment applications. Moreover, the inherent static knowledge cutoff point and the time misalignment defects of the model fail to predict the trend of the latest events. More importantly, they not only may generate imperceptible inferential illusions but also construct false associations in causal attribution, thus diluting the reliability of prediction results. Besides, the high sensitivity of the prompt mechanism, insufficient dynamic modeling capabilities, and the black box characteristic of the decision-making process collectively lead to significant deficiencies in stability, reproducibility, and interpretability.

4 Datasets

In this section, we comprehensively summarize the typical, representative and public available social media dataset including text modality, image modality, video modality, and metadata information, and provide their description and statistics in detail, as shown the Table 5.

4.1 Dataset overview

4.1.1 SMPD-image

The social media popularity image dataset (SMPD-Image) collected from Flickr, a large-scale social image dataset, is provided by the ACM Multimedia Challenge (Wu et al. 2019), which consists of 486K social media posts from 70k users. Each post comprises multi-type metadata information, such as the high-resolution image, category, lab, description, time, temporal-spatial information, additional information, user profiles, etc.

4.1.2 SMPD-video

The social media popularity video data set (SMPD-Video) collected from Tiktok is a large short-form social video data set, containing 6K videos and 4.5K users (Ye et al. 2025b) with metadata information, such as sharing records, user profiles, image, text, time, location, and category, etc.

4.1.3 TCIP2017

The public temporal popularity image collection (TCIP) data set (Wu et al. 2017) obtained from Flickr is a large-scale social media popularity prediction dataset. This dataset

Table 5 Public available social media datasets

Reference	Dataset	URL	Referenced	Accessed Time
Wu et al. (2019)	SMPD-Image	http://smp-challenge.com/dataset	Xu et al. (2025b); Hu et al. (2024); Cheng et al. (2024); Cho et al. (2024); Wang and Zhang (2017); Lai et al. (2020); He et al. (2019); Kang et al. (2019); Lv et al. (2017); Huang et al. (2018); Wang et al. (2020); Hidayati et al. (2017); Huang et al. (2017); Lin et al. (2019); Hsu et al. (2022); Zhang et al. (2023); Lin and Lee (2024); Zhang et al. (2024b); Xu et al. (2020); Tan et al. (2022); Hsu et al. (2019); Chen et al. (2019b); Wu et al. (2022); Mao et al. (2023); Liu et al. (2023b, c); Hsu et al. (2023); Chen et al. (2023); Tu et al. (2024); Ye et al. (2025a, b); Ding et al. (2019); Lin et al. (2021); Chen et al. (2022); Zhang et al. (2018a); Wang et al. (2023); Zhang et al. (2024a); Xu et al. (2025a); Chen et al. (2024)	January 23, 2026
Ye et al. (2025b)	SMPD-Video	http://smp-challenge.com/dataset	Ye et al. (2025b)	January 23, 2026
Wu et al. (2017)	TCIP2017	https://github.com/social-media-prediction/flickr-data-prediction-2017	Cheng et al. (2024); Wu et al. (2017); Wang et al. (2023)	January 23, 2026
Ni et al. (2025)	MicroLens	https://github.com/westlake-repl/MicroLens	Zhong et al. (2024); Cheng et al. (2025); Ni et al. (2025)	January 23, 2026
Kim et al. (2020)	Instagram	https://sites.google.com/site/sbk-imcv/dataset	Purba et al. (2020); Ziyada and Shamoi (2024); Carta et al. (2020); Zhang et al. (2018b); Ji et al. (2023); Xu et al. (2025a)	January 23, 2026
Wu et al. (2018)	YouTube	https://github.com/avalanc-hesiqi/youtube-engagement	Kim et al. (2024); Li et al. (2016); Wang and Zhang (2017); Halim et al. (2022); Amrutha (2022); Sangwan and Bhatnagar (2021, 2025)	January 23, 2026
Jing et al. (2017)	Micro-Videos	http://acmmm2016.wixsite.com/micro-videos	Jing et al. (2017); Su et al. (2020)	January 23, 2026
Cao et al. (2017)	Weibo	https://github.com/CaoQi92/DeepHawkes	Zhang et al. (2021); Guan et al. (2025)	January 23, 2026
Rogoz et al. (2024)	GPVD	https://github.com/ana-rogoz/PoPreRo	Rogoz et al. (2024); Kayal et al. (2025)	January 23, 2026

includes 680K social media posts with image, each post comprises image, user profiles, time, label, and metadata information, etc.

4.1.4 MicroLens

The MicroLens data set released by Ni et al. (2025) contains the micro-form video data set on two different scales, which is a comprehensive and diverse micro-video recommendation data set, such as the MicroLens 50K users dataset and the MicroLens 100K users dataset. This dataset is associated with multimodal content, including the interaction behaviors between user and item, one million users, and 91 thousand full-length micro-videos. Each sample has interaction behaviors, titles, cover images, raw text, audio, video, and video comments, etc.

4.1.5 Instagram

Kim et al. (2020) publish the Instagram influencer dataset, which contains 33,935 influencers (e.g., account, biography, number of followers) and 10,180,500 posts (e.g., captions, number of likes, comments). Each influencer includes a category label, such as beauty, family, fashion, fitness, food, interior, pet, travel, etc.

4.1.6 YouTube

The YouTube video data set contains 5,331,204 videos from 1,257,412 channels between July 1st and August 31, 2016 Wu et al. (2018). Each video has at least 100 views within 30 days after upload, associated with the metadata information, such as video id, title, description, time, category, duration, definition, channel id, channel title, topic ids, etc.

4.1.7 Micro-videos

The dataset is provided by the lab for Media Search at the National University of Singapore from July 2015 to October 2015 Jing et al. (2017), which contains 303,242 user generated micro-videos with 98,166 users, 120,324 following relationships, and 1.6 million video posts collected from the online micro-video sharing site Vine. All micro-videos are less than 8 seconds, and others is about 6-7 seconds.

4.1.8 Weibo

The Weibo data set released by Cao et al. (2017) contains 119,313 messages on June 1, 2016. Each information item represents a certain message with the metadata information, such as message_id, user_id, root_user_id, publish_time, and retweets record, etc.

4.1.9 GPVD

The global popular video dataset (GPVD) sources from YouTube (Kayal et al. 2025), which comprises the top 50 trending video daily in 109 countries over 589 days (From February 13, 2021 to March 17,2023). This dataset with broader geographic coverage contains approximately 5,450 observations per day, totaling 3,210,050 observations (1,302,698 unique videos) over two years. Each video has a video id, trending countries, metadata information, etc.

4.1.10 PoPreRo

The PoPreRo data set collected from Reddit comes from five different Romanian sub reddit channels (Rogoz et al. 2024), which involve 28,107 samples (e.g., 14,289 obscure samples and 13,818 popular samples). Each sample is associated with metadata information, such as title, content, number of comments, number of up and down votes.

4.2 Model comparison models and datasets

In this subsection, we primarily explore various representative methods for social media popularity prediction on the SMPD and ICIP data set, and compare the state of the art model performance with other models, as shown in the Table 6.

From Table 6, we obviously find that the SKAPP model displays the superior performance for social media popularity prediction on the SMPD, as evaluated by the Mean Squared Error (MSE) indicator, which outperforms several the state-of-the-art methods, achieving satisfactory results, such as HyperFusion, HVLA, SMP-LLM, MultiSmpLLM, THGFormer, RAGTrans, PPCL, etc. Meanwhile, the SKAPP also obtain the best performance on the ICIP, as measured by Mean Squared Error (MSE) and Mean Absolute Error (MAE) indicator. It is worth noting that the SKAPP model realizes the excellent performance in user generated content analysis, which not only shows its powerful ability of multimodal feature extraction and learning capabilities, but also demonstrates the generalization ability of the SKAPP model. However, we fail to reveal the model performance under the classification indicators due to the self-built dataset, few methods, and the distributed evaluation indicators, but the AMPS model provided by Cho et al. (2024) is the state-of-the-art methods on the SMPD data set, such as the accuracy, precision, Recall, F1 are 0.765, 0.763,0.833, and 0.797 for the unpopular items, and are 0.765,0.766,0.68,0.721 for the popular items. Recently, with the powerful semantic understanding and logical reasoning abilities of large language models, Wang et al. (2024) and Chen et al. (2024) propose the representative MultiSmpLLM and SMP-LLM models, but do not achieve the best performance, which can benefit from the prompt quality and domain-specific details.

5 Evaluation metrics

In this section, according to the two task formulations defined in Section 2.2, we mainly outline the evaluation metrics to evaluate the performance of the social media popularity prediction method, including the classification indicators and the regression indicators. The choice of metrics depends on whether the goal is to predict popularity levels or popularity scores.

5.1 Classification metrics

Classification metrics are often used to predict popularity levels in this paper, including $Precision$, $Recall$, $F1$, and $Accuracy$. Table 7 summarizes their formulations and representative references. These metrics are standard in the field, and the detailed derivations can be found in cited references.

5.2 Regression metrics

These metrics are widely adopted to predict the popularity values, which measure the deviation between predicted and actual popularity scores, such as the Root Mean Square Error

Table 6 Performance comparison of representative models On SMPD and ICIP dataset

Reference	Model	Years	SMPD			ICIP		
			MSE	MAE	SRC	MSE	MAE	SRC
Lv et al. (2017)	-	2017	2.4351	1.1354	0.7773	-	-	-
Li et al. (2017)	-	2017	2.7293	1.2475	0.7406	-	-	-
Huang et al. (2017)	-	2017	1.15	0.75	0.88	-	-	-
Hidayati et al. (2017)	-	2017	0.977	0.718	0.898	-	-	-
Wang and Zhang (2017)	-	2017	0.8793	-	-	-	-	-
Zhang et al. (2018a)	UHAN	2018	2.246	1.13	-	-	-	-
Huang et al. (2018)	SMHP	2018	-	0.596	0.912	-	-	-
He et al. (2019)	-	2019	-	1.298	0.72	-	-	-
Kang et al. (2019)	-	2019	2.074	1.248	0.748	-	-	-
Lin et al. (2019)	LDS	2019	2.011	1.038	0.835	-	-	-
Hsu et al. (2019)	-	2019	3.561	1.487	0.656	-	-	-
Chen et al. (2019b)	-	2019	-	1.195	0.763	-	-	-
Ding et al. (2019)	DNN	2019	1.432	3.361	0.649	-	-	-
Lai et al. (2020)	Hyfea	2020	-	1.383	0.702	-	-	-
Abousaleh et al. (2020)	VSCNN	2020	0.97	0.73	0.9014	-	-	-
Xu et al. (2020)	FE	2020	-	1.401	0.636	-	-	-
Lin et al. (2021)	-	2021	-	1.61	0.58	-	-	-
Tan et al. (2022)	-	2022	-	1.4249	0.6928	-	-	-
Wu et al. (2022)	DMfirstTry	2022	-	1.24	0.71	-	-	-
Chen et al. (2022)	-	2022	-	1.22	0.75	-	-	-
Zhang et al. (2023)	DSN	2023	-	1.192	0.763	-	-	-
Wang et al. (2023)	-	2023	2.477	1.237	0.742	1.651	1.736	0.72
Mao et al. (2023)	-	2023	-	1.094	0.768	-	-	-
Liu et al. (2023b)	SMPC	2023	-	1.1728	0.7908	-	-	-
Liu et al. (2023c)	SGG	2023	2.63	1.142	0.813	-	-	-
Hsu et al. (2023)	-	2023	-	1.241	0.718	-	-	-
Chen et al. (2023)	DFT-MOVLT	2023	-	1.17	0.753	-	-	-
Zhong et al. (2024)	PSSA	2024	1.32	0.738	0.891	-	-	-
Lin and Lee (2024)	MMF	2024	-	0.92	0.87	-	-	-
Zhang et al. (2024b)	PPCL	2024	2.374	1.164	0.762	-	-	-
Hu et al. (2024)	DSPT	2024	-	1.1543	0.752	-	-	-
Cheng et al. (2024)	RAGTrans	2024	3.2763	1.3396	0.5859	1.2351	0.7149	0.5914
Zhang et al. (2024a)	THGFormer	2024	3.5673	1.4138	0.5741	1.5615	0.8326	0.4759
Wang et al. (2024)	MultiSmpLLM	2024	-	1.02	0.89	-	-	-
Chen et al. (2024)	SMP-LLM	2024	1.4223	0.6432	0.8866	-	-	-
Tu et al. (2024)	HVLA	2024	-	1.197	0.7341	-	-	-
Xu et al. (2025b)	-	2025	2.46	1.172	0.792	-	-	-
Ye et al. (2025a)	HyperFusion	2025	-	1.2402	0.7324	-	-	-
Xu et al. (2025a)	SKAPP	2025	1.18196	0.8249	0.8414	0.9662	0.6367	0.6965

($RMSE$), Mean Absolute Error (MAE), R^2, Spearman Rank Correlation (SRC), Mean Squared Error (MSE), Mean Absolute Percentage Error ($MAPE$), Mean Squared Logarithmic Error ($MSLE$), Normalized Mean Squared Error ($nMSE$). Table 8 provides detailed definitions and key references. These metrics are standard in this field, and the detailed derivations can be found in cited references.

Table 7 Summary of the classification indicators. (TP, TN, FP, FN represents true cases, true negative cases, false positive cases, and false negative cases, respectively)

Formulation	Reference
$Precision = \frac{TP}{TP+FP}$	Kim et al. (2024); Cho et al. (2024); Qian et al. (2022); Nisa et al. (2021); Ziyada and Shamoi (2024); Zhu et al. (2022); Liu et al. (2023a); Liao et al. (2019); Zhang et al. (2018b); Arazzi et al. (2023); Rogoz et al. (2024); Saeed et al. (2022); Prado-Romero et al. (2020)
$Recall = \frac{TP}{TP+FN}$	Kim et al. (2024); Cho et al. (2024); Qian et al. (2022); Ding et al. (2018); Nisa et al. (2021); Ziyada and Shamoi (2024); Zhu et al. (2022); Carta et al. (2020); Liu et al. (2023a); Zhang et al. (2018b); Arazzi et al. (2023); Rogoz et al. (2024); Kayal et al. (2025); Saeed et al. (2022)
$F1 = \frac{2*Precision*Recall}{Precision+Recall}$	Kim et al. (2024); Cho et al. (2024); Qian et al. (2022); Ding et al. (2018); Li et al. (2016); Jeon et al. (2020); Nisa et al. (2021); Ziyada and Shamoi (2024); Zhu et al. (2022); Carta et al. (2020); Liu et al. (2023a); Liao et al. (2019); Zhang et al. (2018b); Arazzi et al. (2023); Rogoz et al. (2024); Saeed et al. (2022); Prado-Romero et al. (2020)
$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$	Kim et al. (2024); Cho et al. (2024); Ziyada and Shamoi (2024); Arazzi et al. (2023); Liu et al. (2023a); Weissburg et al. (2022); Tang et al. (2022); Nisa et al. (2021); Carta et al. (2020); Bielski and Trzcinski (2018); Zhang et al. (2018b)

6 Applications

In this section, we primarily introduce the application of social media popularity prediction, which has been widely adopted as a enabling technology across several downstream real-world domains, such as recommendation systems, hot discovery, marketing management, content optimization, and public opinion monitoring. For each domain, we highlight how popularity prediction contributes to core objectives, as shown the Fig. 6.

6.1 Recommendation systems

Recommendation systems aim mainly to present relevant content to users based on their historical preferences learned from social media content through adaptive algorithms, which gradually become a key supporting technology (Li et al. 2024b). Social media popularity prediction improves this process by incorporating predictive popularity as an auxiliary ranking signal (Ni et al. 2025). Specifically, systems first estimate the potential future engagement of user-generated content (UGC), then match high-popularity items with target user groups based on profile similarity. This recommendation method not only provides high quality content to quickly reach precise users, but also enhances user browsing experience and the retention time on social platform, thus optimizing traffic distribution efficiency.

6.2 Hot discovery

User generated content explosively grows on social platforms daily, in which a small fraction of information attracts widespread attention from many users. Social media popularity prediction plays an forward-looking core role in the field of hot topic discovery (Liu 2025), it not only tremendously promotes the paradigm shift of the hot topic from passive capture to active prediction, but also greatly innovates the traditional post event inductive hotspot tracking model with a forward-looking monitoring framework. Concretely, social media popularity prediction models accurately identify and evaluate potential hotspots in advance by analyzing multimodal real-time features. This capability enables platforms and researchers to accurately assess the explosive potential of emerging topics and sudden public events, fundamentally innovating traditional post-event hotspot tracking.

6.3 Marketing management

In marketing management, social media popularity prediction provides scientific decision support across the entire campaign lifecycle through multidimensional analysis of business information (Verma et al. 2025). For example, according to user preferences for marketing content, companies choose the appropriate time and channel to improve the accuracy and conversion efficiency of the advertising reach,

Table 8 Summary of the regression indicators. ($\hat{y}_i$ is the predictive value, y_i is the actual value, and n is the sample size)

Metrics	Formulation	Reference
R^2	$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$	Gupta and Yang (2019); Yang et al. (2024); Sangwan and Bhatnagar (2025)
Root Mean Square Error	$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}$	Li et al. (2024a); Lv et al. (2017); Chen et al. (2023); Wu et al. (2020)
Mean Absolute Error	$MAE = \frac{1}{n}\sum_{i=1}^{n}\left\|y_i - \hat{y}_i\right\|$	Xu et al. (2025b); Hu et al. (2024); Cheng et al. (2024); Zhang et al. (2021); Abousaleh et al. (2020); Tang et al. (2022); Wu et al. (2017); Lai et al. (2020); Li et al. (2018); He et al. (2019); Gupta and Yang (2019); Kang et al. (2019); Lv et al. (2017); Huang et al. (2018); Wang et al. (2020); Hidayati et al. (2017); Huang et al. (2017); Lin et al. (2019); Hsu et al. (2022); Nguyen et al. (2019); Sangwan and Bhatnagar (2021); Yang et al. (2024); Sangwan and Bhatnagar (2025); Zhong et al. (2024); Xu et al. (2020); Tan et al. (2022); Li et al. (2017); Hsu et al. (2019); Chen et al. (2019b); Wu et al. (2022); Mao et al. (2023); Liu et al. (2023c); Hsu et al. (2023); Chen et al. (2023); Tu et al. (2024); Ye et al. (2025a); Ding et al. (2019); Lin et al. (2021); Chen et al. (2022); Cheng et al. (2025); Zhang et al. (2024a); Xu et al. (2025a); Wang et al. (2024); Chen et al. (2024); Wu et al. (2019)
Spearman Rank Correlation	$\rho = 1 - \frac{\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$	Xu et al. (2025b); Hu et al. (2024); Cheng et al. (2024); Zhang et al. (2021); Abousaleh et al. (2020); Li et al. (2024a); Wu et al. (2017); Lai et al. (2020); Li et al. (2018); He et al. (2019); Kang et al. (2019); Lv et al. (2017); Wang et al. (2020); Huang et al. (2018); Hidayati et al. (2017); Huang et al. (2017); Lin et al. (2019); Hsu et al. (2022); Zhong et al. (2024); Zhang et al. (2023); Tan et al. (2022); Xu et al. (2020); Aloufi et al. (2017); Hsu et al. (2019); Chen et al. (2019b); Su et al. (2020); Wu et al. (2022); Mao et al. (2023); Liu et al. (2023c); Hsu et al. (2023); Chen et al. (2023); Tu et al. (2024); Ye et al. (2025a); Ding et al. (2019); Lin et al. (2021); Chen et al. (2022); Bielski and Trzcinski (2018); Cheng et al. (2025); Zhang et al. (2024a); Xu et al. (2025a); Wang et al. (2024); Chen et al. (2024); Wu et al. (2019, 2023); Zhu et al. (2020)
Mean Squared Error	$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2$	Xu et al. (2025b); Cheng et al. (2024); Abousaleh et al. (2020); Wang and Zhang (2017); Tang et al. (2022); Kang et al. (2019); Lv et al. (2017); Wang et al. (2020); Hidayati et al. (2017); Huang et al. (2017); Lin et al. (2019); Hsu et al. (2022); Li et al. (2017); Hsu et al. (2019); Liu et al. (2023c); Ding et al. (2019); Zhang et al. (2018a, 2024a); Xu et al. (2025a); Chen et al. (2024)
Mean Absolute Percentage Error	$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left\|\frac{\hat{y}_i - y_i}{y_i}\right\|$	Zhang and Bao (2020); Yang et al. (2024); Ye et al. (2025b)
Mean Squared Logarithmic Error	$MSLE = \frac{1}{n}\sum_{i=1}^{n}\left(log\left(1 + y_i\right) - log\left(1 + \hat{y}_i\right)\right)^2$	Ye et al. (2025a)
Normalized Mean Squared Error	$nMSE = \frac{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}$	Jing et al. (2017); Cheng et al. (2025)

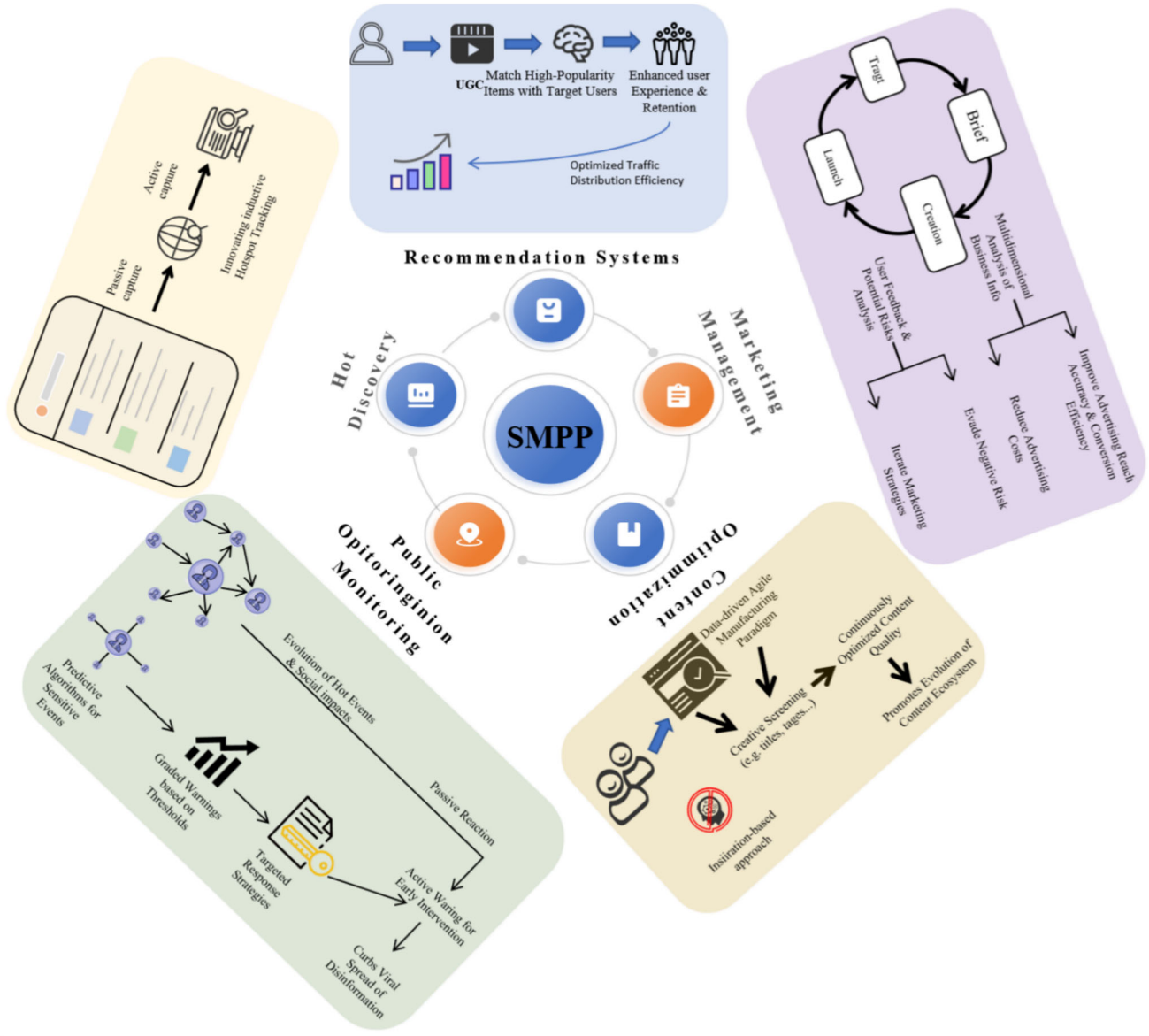


Fig. 6 The Application of social media popularity prediction

further reducing advertising costs. Meanwhile, based on user feedback on marketing content and potential public opinion risks, companies optimize marketing strategies in a timely manner and evade negative risk, providing iteration support for subsequent marketing.

6.4 Content optimization

Traditionally, content creation relied on inspiration and experience. However, The advent of social media popularity prediction has shifted this logic toward a data-driven, agile manufacturing paradigm. Meanwhile, the quality of social media content is continuously optimized in real time based on its popularity, which not only significantly improves its dissemination efficiency but also promotes the evolution of the entire content ecosystem towards being more accurate and efficient (Chaipornkaew and Banditwattanawong 2025). For example, the user obtains the potential impact of different solutions by inputting the preliminary creative ideas (e.g., titles, covers, keywords, and topic tags) into the predictive models in the early stage, thereby achieving data-driven creative screening.

6.5 Public opinion monitoring

Public opinion governance requires timely intervention in sensitive events and disinformation campaigns. SMPP offers a feasible solution by predicting the evolution of hot events

and their potential social impacts using dedicated algorithms (Romaniuk et al. 2025). When key risk indicators—such as interaction rate, emotional polarization index, or cross-community dissemination speed—exceed predefined thresholds, the system issues graded warnings and recommends targeted response strategies. This approach transforms governance from passive reaction to active warning, significantly improving the timeliness and accuracy of monitoring, reducing costs of social trust, and curbing viral spread of disinformation through early intervention.

7 Challenges and opportunity

In this section, we comprehensively discuss the future research directions of social media popularity prediction, which is considered as an interdisciplinary field. In addition to improving predictive accuracy, we should tackle and take advantage of the following challenges and opportunities in future research work.

7.1 Diffusion structure

Currently, social media popularity prediction research mainly focuses on social media content without considering the underlying topological structure of information dissemination (Cheng et al. 2025; Zhang et al. 2024a), which does not explain the difference between the same social media content on different social networks with topological structure. In fact, user generated content is forwarded and shared on social networks, thus the topological structure of different social networks results in different efficiency and paths of information diffusion (Liu et al. 2025c). For example, social media content achieves viral spreading through a few retweets on social networks with a highly connected and short path structure, leading to an exponential trend of popularity. However, social media content slowly spreads along the lengthy propagation paths step by step on social networks with low connectivity and long path. Therefore, combining the structure of social networks and social media content is a crucial research direction. In response to this challenge, future research on social media popularity constructs a structure and content fusion model, which learns the information propagation process with topological structure and the multimodal content semantics of user generated content with the graph structured neural networks and pre-trained models, and aggregates them through the fusion mechanisms. Meanwhile, we explore the contributions of structure and content through decoupling learning mechanisms, demonstrating the differential evolution of the same content on different social networks.

7.2 Model interpretability

With its powerful abstract learning ability of deep learning, researchers propose many deep learning methods and large language models methods for predicting social media popularity, which gradually become the mainstream solution. Although existing work has achieved promising results in predictive accuracy and generalization ability through automatically learning complex patterns and nonlinear correlations of multimodal content (Weissburg et al. 2022). However, insufficient model interpretability is the core limitation, which lies in highly opaque inference and the analysis mechanism within the model, which significantly exhibits its black box characteristics (ŞAHiN et al. 2025). To address the above concerns, we explicitly track the causal relationship from multimodal content features for social media popularity prediction by introducing counterfactual reasoning or attention attribution mechanisms. Meanwhile, we decouple the hidden representations of the proposed model with a structured knowledge graph, revealing which content modalities or patterns mainly affect the predictive results.

7.3 Semantic alignment

User generated content is inherently composed of various types of multi-modal content, such as text, image, and video, etc. Existing efforts directly extract multi-modal features from social media content with the pre-trained models or the self trained models. Then, the extracted features are simply concatenated or the cross fusion learning using the attention mechanisms for popularity prediction under the semantic alignment of multi-modal situation (Zou et al. 2025). In fact, due to various improper operations during the data collection process, the dataset inevitably results in noise and data loss, which leads to the semantic loss and the semantic misalignment of multi-modal features in the feature learning process, which poses challenges for accurately predicting social media content (Song et al. 2025). To overcome this issue, we should design a contrastive alignment and missing completion to address multi-modal semantic loss and misplacement problem in future work. We can force semantic alignment under noise factors by introducing cross modal contrastive learning frameworks, and explicitly reconstruct or mask representation correction of missing modalities with the variational autoencoder, alleviating the semantic loss caused by data collection process.

7.4 Data augmentation

Users generate a large amount of information on the social network every day. However, only a small amount of

information items have garnered significant attention from many users, and most of the content only receives a small amount of attention. Therefore, the collected social media dataset severely suffers from data sparsity and uneven distribution, which not only limits the predictive performance of the model but also weakens its robustness (Sapkota et al. 2025a). Data augmentation techniques are considered as a key solution, which produces semantically coherent and communicative social media content effectively expanding the training samples for rare events with artificial intelligence generates content (Sapkota et al. 2025b), such as sudden public events, alleviating the imbalance and data sparsity. To overcome this barrier, we should build data augmentation framework based on generative artificial intelligence, which semantically expands and resamples rare samples with the multimodal generation mechanism, balancing data distribution and improving the model robustness in long tail scenarios.

8 Discussion

In this section, we will provide a systematic comparison and trade-offs of the five categories of methods reviewed in Section 3 in the data requirements, computational cost, semantic awareness, and interpretability characteristics, etc. which are essential for both selecting appropriate methods for social media popularity prediction tasks and identifying future research directions. Meanwhile, we outline the methodological comparison.

According to different information sources and application scenarios, each methodological family makes different assumptions about what drives social media popularity. For example, generative methods consider that social media popularity evolves through a self-exciting mechanism where historical events increase the instantaneous probability of new events occurrence, which mainly rely on temporal interaction sequences rather than social media content, and thus they are uniquely suited for understanding viral dynamics but are blind to semantic content. Meanwhile, feature engineering methods assume that social media popularity prediction can be obtained by combining traditional machine learning methods and handcrafted features derived from social media content, which are highly dependent on expert knowledge. Their strength lies in efficiency and interpretability, but they cannot automatically discover complex patterns. Furthermore, time series methods rely on the fact that social media popularity follows predictable temporal patterns (e.g., trends, seasonality, and autocorrelation) extrapolated from historical numerical sequences, but they exclusively ignore the semantics of social media content. Moreover, multimodal methods following the "feature extraction and shallow learning" paradigm to capture the rich semantic information from multimodel and then fed into traditional machine learning predictors, inheriting the semantic understanding of the pre-trained models while maintaining the efficiency of traditional machine learning, and may easily trapped in information bottleneck due to the frozen feature extractor. Compared to multimodal machine learning methods, multimodal deep learning methods rely on large-scale labeled data and substantial computational resources to train deep networks to learn the nonlinear, and the trade-off is reduced model interpretability and increased overfitting risk. Besides, Large language model methods apply the contextual under-

Table 9 Comprehensive comparison of the reviewed methods

	Generative Methods	Feature Engineering Methods	Time Series Methods	Multimodal Methods		Large Language Model Methods
				Machine Learning	Deep Learning	
Data type	Event timestamps	Handcrafted features	Historical numerical sequences	Multimodal content	Multimodal content	Multimodal content
Cost	Lower	Low	Lower	Low–Medium	High	Higher
Interpretability	High	High	High	Medium	Low	Lower
Semantic Awareness	None	Shallow	None	Deep	Deep	Deeper
Handling Features	Not applicable	Manual features	Not applicable	Automatic extraction	Automatic extraction	Automatic extraction reasoning
Cold-Start	Poor	Poor	Poor	Poor–Medium	Poor	Good
Real Time	Excellent	Excellent	Excellent	Good	Poor	Poor
Scenarios	Modeling of viral transmission	Engineering based rapid deployment	Regular popularity prediction	Multi modal content understanding		Semantic complexity

standing and few-shot reasoning capabilities of LLMs to predict social media popularity but their strength is semantic depth and cold-start capability, but they suffer from high latency, static knowledge cutoffs, and hallucination risks.

Table 9 shows a unified comparison in several key dimensions, which extends the method-specific observations of Section 3.1 to Section 3.5 by introducing comparative metrics. We can observe that the generative methods, features engineering methods and time series methods are low in terms of computational cost, semantic awareness and cold-start. Despite the poor real time, interpretability and efficiency, multimodal methods and large language model methods can analyze more complex semantic features of social media content, achieving satisfactory results.

9 Conclusion

In this paper, we comprehensively review the existing methods of social media popularity over the past decade. According to feature extraction and learning strategies, we propose a novel taxonomy to divide existing works into five classes and their sub-classes, and analyze their strengths and weaknesses. To better understand the task of predicting social media popularity, we introduce the basic concepts related to social media content and task definition, which includes the classification task and the regression task. Meanwhile, we provide the resource compilation, which contains the representative model performance, public dataset, and evaluation indicators. Besides, we discuss the application and potential directions for future research. This paper aims to help readers quickly understand the latest development of social media popularity prediction.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval Not applicable.

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